5.2.2 Matter: hot or cold

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
	(i) ratios of numbers of particles in quantum states of different energy, at different temperatures (classical approximation only)	*M0.3*
	(ii) qualitative effects of temperature in processes with an activation energy (for example, changes of state, thermionic emission, ionisation, conduction in semiconductors, viscous flow).	HSW7, 8
(b)	*Make appropriate use of, by sketching and interpreting:*	
	(i) graphs showing the variation of the Boltzmann factor with energy and temperature.	*M3.12*
(c)	*Make calculations and estimates of:*	
	(i) ratios of characteristic energies to the energy kT	*M0.3*